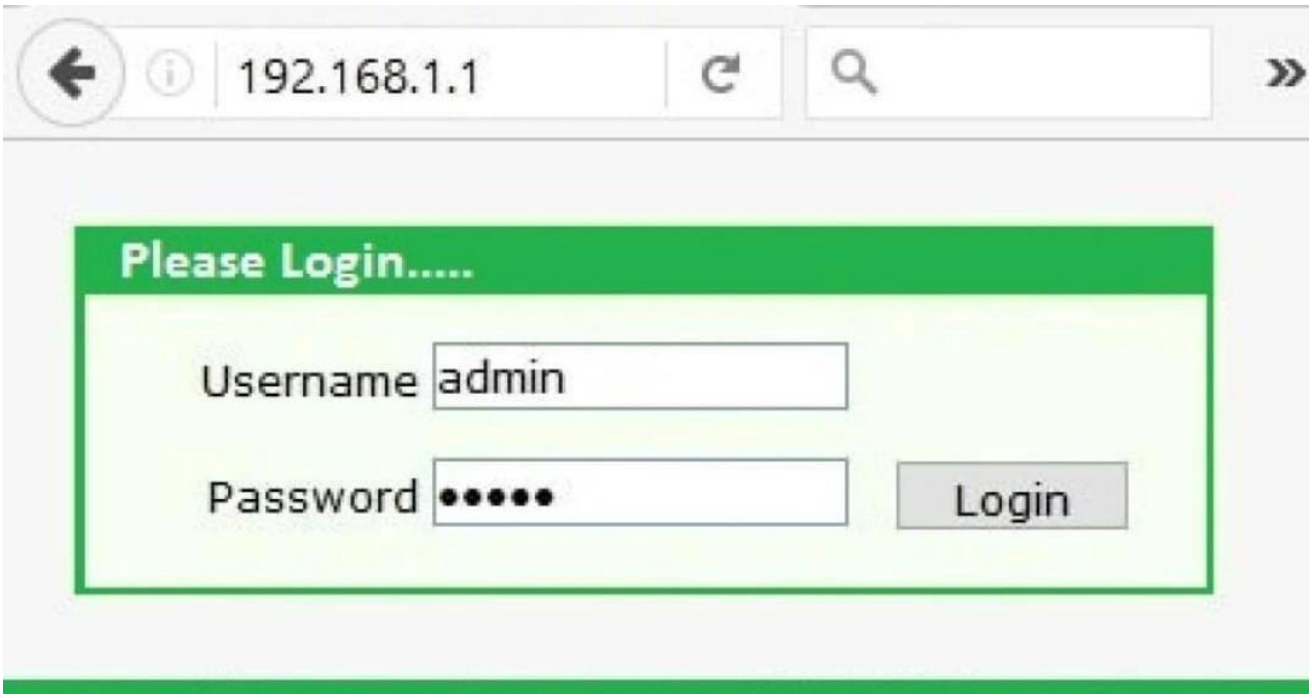
UNDERSTANDING AND MAINTENANCE OF NETWORKS

Practical Assignment

Configuring Basic Network Hardware

Q1. Log in to your home Wi-Fi router's admin panel and take a screenshot of the login page.

First, I connected my laptop or mobile to my home Wi-Fi network. Then I opened a web browser and entered my router's IP address (for example, 192.168.1.1). The router login page appeared. I entered the username and password to open the admin panel.



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Q2. Note the IP address, subnet mask, and default gateway of your device.

I checked the network details of my device while it was connected to Wi-Fi. The details are:

- IP Address: 192.168.1.100
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

These details help my device communicate with the router and connect to the internet.

Q3. Change the device's IP setting from DHCP to Static IP and test the internet connection.

First, my device was set to obtain an IP address automatically (DHCP). For the test, I changed it to a static IP address in the same network range as my router.

- Static IP Address: 192.168.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS:8.8.8.8.

After saving the settings, I checked whether the internet was working. My test result was: _____.

After testing, I changed the IP setting back to automatic (DHCP) to avoid network problems.

Q4. Find the DHCP range in your router and explain what happens if two devices get the same IP address.

The DHCP range is the range of IP addresses that a router automatically assigns to connected devices.

Current DHCP Range: 192.168.1.100 – 192.168.1.200

If two devices use the same IP address, an IP conflict can happen. The devices may have trouble communicating with the network, and one or both devices may lose their internet connection.